

For the **desuperheater** shown in Fig. P4.85, liquid water at **state 1** is injected into a stream of superheated vapor entering at **state 2**. As a result, saturated vapor exits at **state 3**. Data for **steady state** operation are shown on the figure. Ignoring stray heat transfer and kinetic and potential energy effects, determine the mass flow rate of the incoming superheated vapor, in kg/min.

